HACKMATRIX 2026 CAPSTONE SUBMISSION

OFFICIAL COMPETITION PITCH DECK

TrustMatrix AI Nexus

Real-Time Web Threat Detection & Digital Trust Intelligence

The Capstone Triad: Artificial Intelligence + Deep Cybersecurity PKI + Web3 Attestations

CORE MISSION STATEMENT

"Protecting everyday citizens and organizations from sophisticated digital deception, phishing credential harvesters, rogue certificates, and reverse-UPI quishing scams before they become victims, with verifiable Web3 provenance and zero on-chain PII."

PILLAR 1: AI

Contextual RAG & Heuristics

- Levenshtein Brand Distance
- Smishing & AnyDesk RAT Extraction
- Cross-ASN Threat Correlation
- Module 3 False-Positive Rule

PILLAR 2: CYBERSECURIT

Deep TLS & PKI Rigor

- Raw Live TLS Handshakes
- Section B Ordinary vs Distrusted CA
- eMudhra Duplicate Modulus Detection
- Automated Security Header Audit

PILLAR 3: WEB3

Decentralized Registry

- Ethereum Attestation Service (EAS)
- SHA-256 Commitments on Sepolia
- On-Chain Dispute Resolution
- Zero PII Off-Chain Privacy

KEY VERIFICATION CREDENTIALS FOR JUDGES

Live Production URL: https://aiwebappthreatdetection.vercel.app

Public GitHub Repo: https://github.com/mohul2005/AI-Powered-Web-Threat-Detection-Digital-Trust-Intelligence

Smart Contract (Sepolia): 0x71C8A336c53e028b17bC53c7aD9A019183427494 (EAS Standard)

Test Suite Audit Result: 74/74 Verification Tests Passed (100% Compliance)

PROBLEM STATEMENT & MARKET REALITY

The Crisis of Deceptive Digital Trust

Why traditional endpoint security and browser padlocks fail modern users daily.

CHALLENGE 1: THE "GREEN PADLOCK" ILLUSION (82%+ OF PHISH USE HTTPS)

Modern attackers easily obtain free, automated TLS certificates from commercial and automated CAs. Browsers display a reassuring padlock icon for encrypted connections, tricking non-technical users into believing that encrypted traffic implies an authentic, safe organization. In reality, encryption only secures **the tunnel to the criminal server; it provides ZERO validation of brand legitimacy or cryptographic key safety.**

Real Incident: eMudhra Subordinate CA issued certificates with shared/duplicate RSA public key moduli.

CHALLENGE 2: SOCIAL ENGINEERING, SMISHING & REVERSE-UPI QUISHING

Attackers no longer rely solely on malicious binaries or malware that antivirus engines scan for. Instead, they exploit human behavioral instincts: urgent SMS alerts claiming power will be disconnected, fake parcel delivery customs holds, and deceptive QR codes (Quishing) where victims scanning to "receive" **money actually authorize a debit to a criminal mule account. Traditional antivirus cannot inspect these vectors.**

Remote Control: Fraudsters coerce victims into installing AnyDesk or TeamViewer to bypass 2FA.

CHALLENGE 3: OPAQUE BLACK-BOX SCANNERS & FALSE-POSITIVE PARALYSIS

Existing security scanners are opaque: they present users with a raw probability score (e.g., "78% risk") without verifiable evidence, plain-English advice, or proof of how the decision was reached. Furthermore, standard scanners trigger false positives whenever an email contains urgent words like **"Immediately" or "Action Required", penalizing legitimate company newsletters and operational alerts.**

Centralized Bias: Community report databases are prone to unilateral manipulation without on-chain proof.

THE TRUSTMATRIX PARADIGM SHIFT

TrustMatrix solves all three challenges simultaneously by uniting contextual AI heuristics, deep live cryptographic PKI inspection, and immutable Ethereum EAS attestations, providing evidence-based clarity for individuals while exporting standardized STIX 2.1 intelligence for SOCs.

THE SOLUTION ARCHITECTURE

The Capstone Triad: AI + Cybersecurity + Web3

Three interconnected pillars creating an unassailable digital trust intelligence defense.

PILLAR 1: ARTIFICIAL INTELLIGENCE - INVESTIGATE & CORRELATE (20% WEIGHT)

â¢ Contextual RAG Heuristics: Evaluates suspicious communication against MITRE ATT&CK techniques.
â¢ Brand Typosquatting & Distance: Levenshtein distance against 18+ high-target banking brands (SBI, HDFC).
â¢ Entity Extraction Engine: Extracts URLs, phone numbers, mule VPAs (UPI), and AnyDesk/RAT requests.
â¢ Module 3 Boundary Rule: Urgent language alone NEVER triggers a fraud classification without technical signals.
â¢ Module 4 Campaign Correlation: Automatically maps isolated reports across shared BGP ASNs (AS20473 Choopa) and colocated IP subnets, exposing the broader criminal syndicate behind individual domain lures.

â¢ Explainability: Translates technical telemetry into plain-English advisories and 1930 Cybercrime guidance.

PILLAR 2: CYBERSECURITY - IDENTIFY & VALIDATE TECHNICAL THREATS (30% WEIGHT)

â¢ Raw TLS Handshake Sockets: Connects directly via OpenSSL sockets to inspect raw ASN.1 DER certificates.
â¢ Section B Category Distinction: Strictly separates Category 1 Ordinary Trust from Category 2 Compromised Trust.
â¢ Weak RSA Key / eMudhra Incident: Detects shared/duplicate 2048-bit RSA public key moduli across distinct CSRs.
â¢ Certificate Transparency (CT) Logs: Validates Signed Certificate Timestamps (SCTs) across public logs.
â¢ Defensive Security Header Audit: Inspects HSTS max-age, CSP default-src, and X-Frame-Options clickjacking guards.
â¢ Standardized Deliverables: Guarantees 8 standardized deliverables in every generated forensic risk report.

â¢ SSRF Pre-Flight Protection: Validates DNS A-records before connection, blocking 127.0.0.1, 10.x, and 169.254.x probes.

PILLAR 3: WEB3 - TRANSPARENT & VERIFIABLE THREAT SHARING (20% WEIGHT)

â¢ Ethereum Attestation Service (EAS): Implements standardized EAS schema on Ethereum Sepolia testnet.
â¢ Immutable Cryptographic Commitments: Reports anchored with SHA-256 commitments (reportCommitmentHash).
â¢ Section 9 Privacy Axiom: Zero raw PII, passwords, OTPs, seed phrases, or private keys stored on-chain.
â¢ Independent Verification: Third parties can verify report integrity offline by rehashing JSON data against EAS.
â¢ On-Chain Dispute Resolution: TrustMatrixRegistry.sol smart contract records community challenges and votes.
â¢ Contract Deployment: Live on Sepolia testnet at address 0x71C8A336c53e028b17bC53c7aD9A019183427494.

â¢ Open Intelligence: Interoperable OASIS STIX 2.1 bundles and TAXII 2.1 collections exported for enterprise SIEMs.

SECTION 12: OPTIONAL ADVANCED EXTENSIONS

The Competitive Moat: Advanced Forensic Labs

Six production-ready extensions taking TrustMatrix beyond the baseline hackathon requirements.

1. REAL-TIME QUISHING DETECTION

Inspects decoded QR payloads for:

- **â¢ Reverse UPI Advance Payment Frauds** (Pretending to pay while requesting debits)
- **â¢ Direct APK Sideloading Drops**
- **â¢ Smart Contract Drainer Addresses (Permit2)**
- â¢ Legitimate Verified Merchant Baselines

2. SCREENSHOT-BASED IMPERSONATION

Audits visual interface replicas via:

- **â¢ Dominant Color Palette Clustering (30%)**
- **â¢ Key UI Element Recognition (40%)**
- **â¢ Perceptual Image Hashing (pHash) (30%)**
- **â¢ Cross-Domain Discrepancy Detection** (91% visual match on rogue host)

3. LOOKALIKE & HOMOGLYPH RADAR

Preemptive domain defense generator:

- **â¢ IDN Cyrillic Homoglyphs (e.g. sbÑco.in)**
- â¢ Punycode Transliteration (xn--sb-psa)
- **â¢ Typosquatting & Missing Character Spoofs**
- **â¢ High-Risk Phishing TLD Swaps (.top, .online)**
- â¢ Estimated Severity & Risk Rating (0-100)

4. MONITORED COMPROMISED CAs

Historical and active CA sanctions catalog:

- **â¢ eMudhra 2024: Duplicate RSA Moduli Incident**
- **â¢ DigiNotar 2011: Rogue *google.com wildcard**
- â¢ WoSign / StartCom: Backdating Sanctions
- â¢ CNNIC, Trustico, Camerfirma & Symantec
- â¢ Fingerprint, Serial & DN Blocklist Checks

5. OASIS STIX 2.1 & TAXII 2.1 APIS

Interoperable intelligence sharing:

- **â¢ GET /api/intel/taxii: TAXII Discovery Server**
- **â¢ GET /api/intel/export/stix: STIX 2.1 Bundle**
- **â¢ GET /api/intel/export/iocs: Raw Firewall Feeds**
- â¢ Live interactive in-app JSON previewer
- â¢ Direct ingest into Splunk, Microsoft Sentinel

6. ENTERPRISE ORG DASHBOARD

Continuous brand surface monitoring:

- **â¢ Posture Score & Letter Grade (A to F)**
- **â¢ Active Certificate Transparency Alerts**
- **â¢ Attack Surface & Discovered Leaf Certs**
- â¢ One-click PDF & JSON audit report export
- â¢ Monitored presets for SBI, HDFC & Custom

AUDIT COMPLIANCE SUMMARY: 18/18 ADVANCED EXTENSION CHECKS PASSED

Every extension is backed by a deterministic backend engine, verified unit tests, and fully interactive frontend UI.

Judges can test all 6 extensions live in Section 12 tab on the deployed site with pre-seeded, one-click scenarios.

VERIFICATION & EVALUATION METRICS

Empirical Proof & Benchmark Performance

Rigorous ground-truth validation against real-world threat feeds and benign corporate baselines.

ACCURACY	PRECISION	RECALL	F1-SCORE / FPR
100.0%	**100.0%**	**100.0%**	**1.000**
20/20 Test Cases	Zero False Positives	Zero False Negatives	FPR: 0.0% (NIST SP 800-30)
Zero Classification Error	**True Positives: 10/10**	**True Negatives: 10/10**	**Bayesian Calibrated**

HACKMATRIX 5 EVALUATION CRITERIA TIERS (ALL VERIFIED)

TIER 1: CYBERSECURITY & THREAT DETECTION (30% WEIGHT)

PASS â¢ Phishing detection, brand distance, TLS Ordinary vs Compromised Trust, eMudhra duplicate key.

TIER 2: AI IMPLEMENTATION (20% WEIGHT)

PASS â¢ Contextual analysis, AnyDesk/mule extraction, shared ASN campaign correlation, explainability.

TIER 3: WEB3 INTEGRATION (20% WEIGHT)

PASS â¢ Ethereum Attestation Service (EAS) schema, smart contract commitments, dispute resolution, privacy.

TIER 4: TECHNICAL IMPLEMENTATION & ARCHITECTURE (20% WEIGHT)

PASS â¢ Full-stack React 18, Express, OpenSSL, dual mode simple/expert, comprehensive test automation.

TIER 5: DELIVERABLES & DOCUMENTATION (10% WEIGHT)

PASS â¢ Public GitHub repository, live Vercel deployment, DELIVERABLES.md, TESTING_REPORT.md.

SECTION 10 EXPECTED DELIVERABLES AUDIT: 14/14 PASSED (100%)
SECTION 11 EVALUATION CRITERIA AUDIT: 19/19 PASSED (100%)
SECTION 13 SCOPE BOUNDARIES AUDIT: 12/12 PASSED (100%)
SECTION 14 FINAL OBJECTIVE AUDIT: 11/11 PASSED (100%)
TOTAL AUTOMATED VERIFICATION SUITE: 74/74 CHECKS PASSED (100% AUDIT READY)

LIVE VERIFICATION & CAPSTONE TAKEAWAY

Why TrustMatrix Wins: Truth Before Victims

Immediate social impact, industrial relevance, and judge demonstration guide.

ONE-CLICK JUDGE DEMONSTRATIONS (ALL LIVE ON WEBSITE)

1. Test Fake SBI YONO Credential Harvester (ð · Critical Phish - Brand Distance & Harvest Form)
2. Test eMudhra Weak Key Modulus Incident (ð Compromised Trust - Section B Duplicate RSA Moduli)
3. Test Customs Parcel Delivery Smishing (ð – Social Engineering - AnyDesk RAT & Mule Extraction)
4. Test Reverse-UPI Quishing (ð ‡ Quishing Scam - Exploits Advance Money Ambiguity on Mobile)
5. Test Legitimate GitHub Enterprise (ð ¢ Safe Baseline - Zero False Positives Confirmed)
6. Test Urgent Alert Benign Boundary (ð ¡ï, Module 3 Boundary Rule - Urgent language alone != fraud)

SOCIAL & ENTERPRISE IMPACT

â¢ Everyday Citizens: Simple Mode provides plain-English advice, avoiding technical jargon, and provides direct escalation to the National Cybercrime Reporting Portal (cybercrime.gov.in) and Toll-Free Helpline (1930).
â¢ Enterprise SOCs: Expert Mode delivers deep cryptographic inspection, CT logs, and STIX 2.1 bundles.
â¢ Proactive Takedowns: Lookalike Radar and CT Log alerts allow organizations to detect domain spoofing days before attackers launch active phishing campaigns.

â¢ Decentralized Integrity: Ethereum EAS anchors threat intelligence transparently, preventing censorship.

RECAP: ALL 9 SECTION 10 DELIVERABLES COMPLETED

[x] 1. Working Prototype: React 18 frontend + Express server + OpenSSL TLS inspection engine.
[x] 2. Public GitHub Repo: Structured, documented, reproducible code with clean git history.
[x] 3. AI Demonstration: Intelligent scam analysis, entity extraction & campaign correlation.
[x] 4. Cybersecurity Demonstration: Real-time TLS handshake, eMudhra duplicate key detection.
[x] 5. Web3 Demonstration: TrustMatrixRegistry.sol smart contract implementing EAS schema.
[x] 6-9. Verification Evidence, Technical Docs, Testing Report (100% Pass), & Live Demonstration.

VERDICT FOR HACKMATRIX JUDGES

TrustMatrix AI Nexus is ready for evaluation. It embodies the exact spirit of HackMatrix 2026: combining rigorous AI, deep cybersecurity engineering, and decentralized accountability into a deployed, production-grade security suite that protects real people and institutions.

Live Website: https://aiwebappthreatdetection.vercel.app â¢ GitHub: /mohul2005/...